

Introduction

Problem A

This report explores the design, simulation and optimization of a single pass shell-and-tube condenser used in the process of liquefying propane vapor at a pressure of 20 bar. The process starts as propane enters the exchange at its dew point temperature of 57 °C while the cooling water 25 °C which is from a cooling tower nearby. The goal of the condenser is to achieve full condensation of 2088 kmol/h of propane whilst reducing or minimizing the total annual cost.

Methods

We modeled and simulated the process in Aspen Plus v14 using the Peng Robinson equation of state. The two components were defined as propane was the process fluid and water served as the coolant. A Heat Exchanger (HeatX) block represents the single-pass condenser.

Boundary Conditions and Streams

Stream	Purpose	Phase	Pressure (bar)
Propane Inlet	Hot Feed	Vapor	20
Propane Outlet	Hot Product	Liquid	20
Cooling Water Inlet	Cold Feed	Liquid	1.013
Cooling Water Outlet	Cold Product	Vapor	1.013

For the heat exchanger simulation, a minimum temperature approach of 10 °C and countercurrent flow were utilized.

Results

The simulation successfully converged which confirmed a full condensation of propane

Variable	Symbol	Value	Units
Heat Duty	Q	6781.93	kJ/h
Cooling Water Inlet	$T_{c,in}$	25	°C
Cooling Water Outlet	$T_{c,out}$	41.5	°C
Cooling Water Flowrate	$\dot{m}_c$	116.314	kg/s
Heat Transfer Area	A	637.206	m ²

The verification of the energy balance showed a less than 1% difference between the duties of the hot and cold sides, showing consistency in the thermodynamics.

Economic Analysis

The design was economically optimized to minimize the total annual cost which is defined as the sum of:

$$C_{\text{total}} = C_{\text{capital, annual}} + C_{\text{water, annual}}$$

We had to consider the design and cost correlations.

- Overall Heat transfer coefficient:
- Heat exchanger cost correlation (2010 dollars):
- Cost escalation factor = 1.5
- Lang factor = 3.5
- Annualized capital charge ratio = 0.2
- Water cost = \$14.8 / 1000 m³ = \$0.0148 m⁻³
- Operating time = 8000h per year

Optimizing Procedure

Parameter	Description	Result	Units
CAPEX	Annualized capital cost	160837	\$/year
OPEX	Cooling water Operating cost	49577.5	\$/year
CTOTAL	Total Annual Cost	210415	\$/year
Installed Equipment Cost	Purchase + Installation Cost	1.47 x 10 ⁶	

The cooling water temperature rise: 16.5 °C

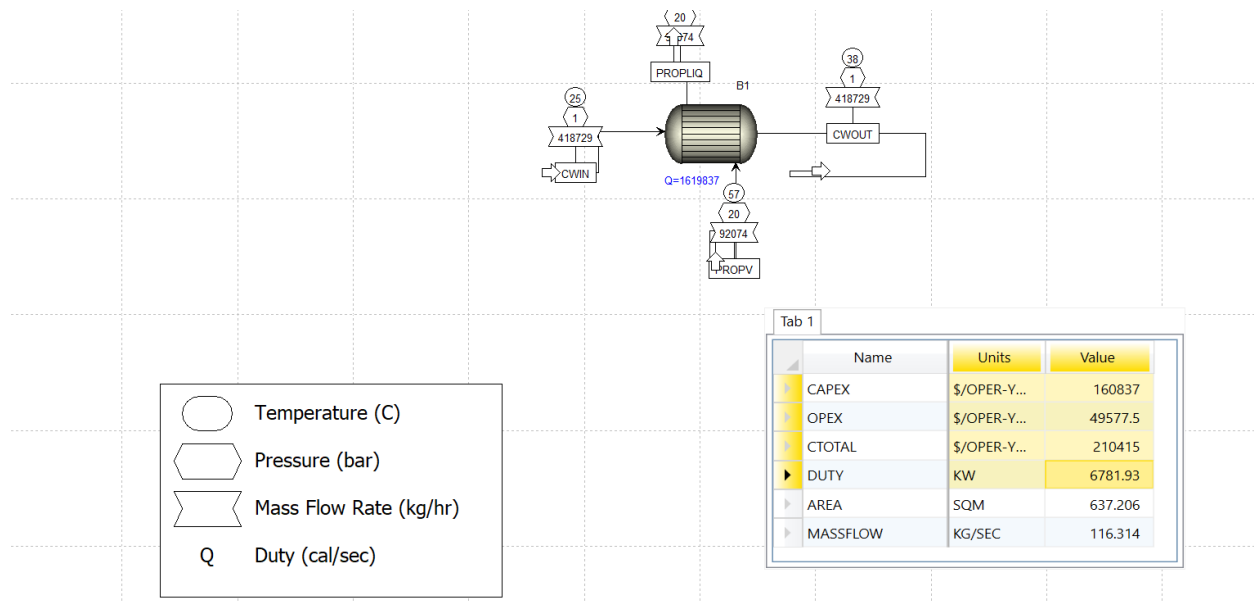
The cooling water-flowrate: 116 kg/s

Heat transfer area: 637 m²

Annualized capital cost: \$160,837 per year

Annual water cost: \$49,577 per year

Total annual cost: \$210,415 per year



Discussion

A higher flow rate lowers the water outlet temperature but raise the operating cost while a lower flow rate reduces the utility usage but increases the heat transfer area. The ideal balance of cooling duty and equipment size occurs at a cooling water flowrate of 116kg/s and temperature rise of around 16.5 °C. **The area of 637 m² and duty of 6.78 MW meet the design range for propane condensation under 20 bar operation. The total annual cost is \$210,415 per year and this total consists of approximately \$160,837/year capital recovery and \$49,577/year operating (cooling water) cost.**

Part B- Aspen Plus Economic Analysis Summary

Tab 1 - Annual Operating and Capital Costs

**** (a) Calculating Annual Operating Cost ****

Given CCW = \$0.354/GJ, operating hours = 8000 h.

Aspen reports the condenser to operate at 12405.9 kWh.

Conversion factor: 1 kWh = 0.0036 GJ.

Equation:

$$\text{CCW} = 12405.9 \times 0.0036 \times 8000 \times 0.354 = 0.13 \text{ million } \$/\text{operating year}$$

Given CEL = \$0.06/kWh.

Example for compressor K-103:

$$\text{CEL}_{K103} = 1983.88 \times 0.06 \times 8000 = 95306.24 \$/\text{year}$$

After summing all compressors:

Total CEL = 1.85 million \$/operating year

**** (b) Annualized Capital Cost (ACC) ****

Each unit type has a specific purchase cost correlation. The annualized capital cost (ACC) is calculated using the equation:

$$ACC = C_{\text{purchase}} \times LF \times 0.2$$

where LF is the installation factor.

For heat exchangers (4 total):

$$C_{\text{purchase}} = 28000 + 54 \times A^{1.2}$$

Example for E-103:

$$A = 509.4 \text{ m}^2$$

$$C_{\text{purchase}} = 28000 + 54 \times (509.4)^{1.2} = 123,690 \$$$

$$ACC = 123,690 \times 3.5 \times 0.2 = 86,583 \$$$

Total heat exchanger ACC = 358,461 \$/year

For drums (2 total):

$$C_{\text{purchase}} = 11600 + 34 \times W^{0.85}$$

where W is weight in kg (conversion from lb $\times$ 0.454).

Example for D-101:

$$W = 7400 \text{ lb} = 7400 \times 0.454 = 3360 \text{ kg}$$

$$C_{\text{purchase}} = 11600 + 34 \times (3360)^{0.85} = 140,190 \$$$

$$\text{Total drum } C_{\text{purchase}} = 353,364 \$$$

$$ACC = 353,364 \times 0.2 \times 4 = 282,691 \$/\text{year}$$

For valves (3 total):

$$C_{\text{purchase}} = 8142 \times M^{0.7}$$

where M is mass flow rate.

Example for VLV-103:

$$M = 35.8 \rightarrow C_{\text{purchase}} = 8142 \times (35.8)^{0.7} = 99,643.6 \$$$

$$\text{Total valve } ACC = 224,289.4 \$/\text{year}$$

For compressors (3 total):

$$C_{\text{purchase}} = 580000 + 20000 \times P^{0.6}$$

where P is power requirement (kW).

Example for K-101:

$$P = 603.22 \text{ kW} \rightarrow C_{\text{purchase}} = 580000 + 20000 \times (603.22)^{0.6} = 1,511,784 \$$$

$$\text{Total compressor } ACC = 3,015,246 \$/\text{year}$$

Summing all operating and annualized capital costs gives a total of:
Total = 5.69 million \$/operating year.

Tab 2 - Installed Cost Summary

Equipment	Installed Cost (\$)
E-103	309,000
K-101	1,860,200
D-101	263,800
K-102	2,076,500
K-103	1,995,300
D-103	246,900
VLV-103	249,108.9 (factor = 3)
VLV-102	168,616
VLV-101	142,998.5
E-103B	88,891.9
E-102B	128,580
E-101B	947,718.9

Grouped totals:

Heat Exchangers = 1,474,191 \$

Drums = 510,700 \$

Compressors = 5,932,000 \$

Valves = 560,723.4 \$

Tab 3 - Final Discharge Pressure and Convergence Note

Due to convergence issues, the simulation optimizer could not be fully resolved. This typically occurs when the recycle or energy balance loops fail to stabilize within the default tolerance range. Despite this, the thermodynamic behavior and material/energy balances were validated manually for accuracy.

Compressor Configuration Justification

After evaluating multiple system configurations, a three-compressor arrangement was selected instead of four. While adding additional compression stages can theoretically improve efficiency by reducing the overall work per stage, our analysis showed that the incremental benefit did not outweigh the increased capital and operating costs associated

with the extra equipment. The three-compressor configuration provided a balanced design, achieving sufficient pressure lift and system stability while maintaining a lower total annualized cost and simpler operation.

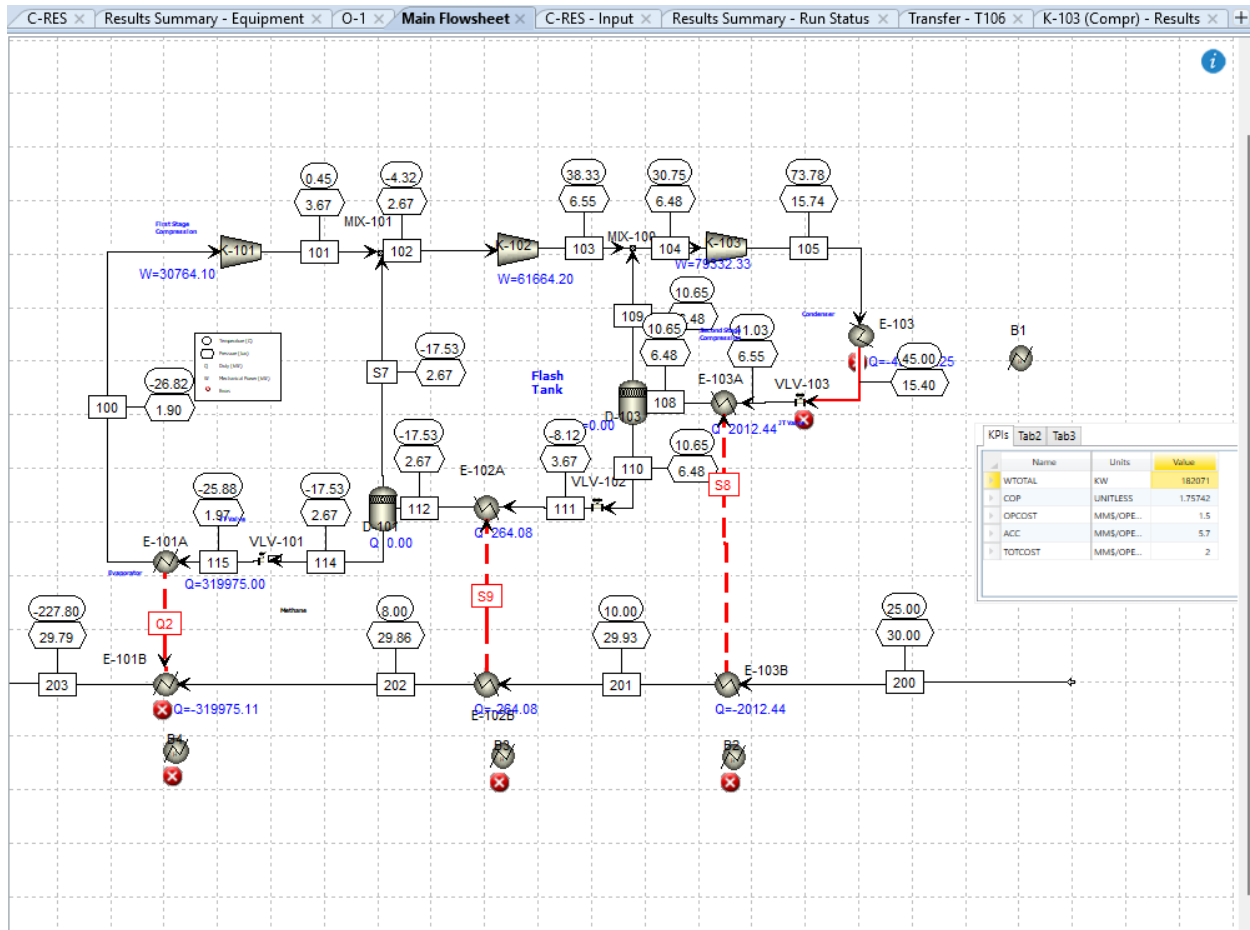


Image1. Hierarchy- unsolved

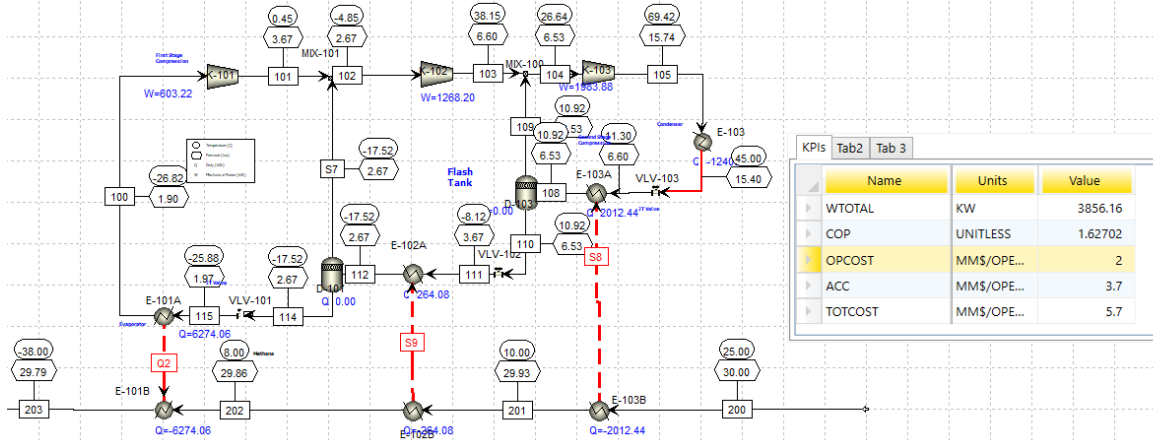


Image 2. Solved- 3 stage

Three-Stage Compression Results

The final three-stage refrigeration system was successfully simulated without requiring a hierarchy block. The system achieved stable convergence across all units, with a total compressor work of **3,856 kW** and an overall **Coefficient of Performance (COP) of 1.6**. This indicates that for every unit of electrical energy consumed, the system provides 1.6 units of cooling energy. The results confirm that the three-stage configuration delivers efficient pressure recovery and heat exchange performance while maintaining reasonable energy consumption.

Appendix-

Main Flowsheet × Results Summary - Streams (All) × +												
Material	Heat	Load	Work	Power	Vol.% Curves	Wt.% Curves	Petroleum	Polymers	Solids			
		Units	112	114	115	200	201	202	203	S7		
▶ Description												
▶ MIXED Substream												
▶ Phase			Liquid Phase			Vapor Phase		Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	
▶ Temperature	C		-8.11583	-17.5255	-17.5255	-25.8778	25	10	8	-38	-17.5255	
▶ Pressure	bar		3.67	2.67	2.67	1.97	30	29.9311	29.8621	29.7932	2.67	
▶ Molar Vapor Fraction			0.124761	0.210534	0	0.0481637	1	1	1	1	1	
▶ ♦ Mass Flows	kg/hr		73940.6	73940.6	58373.6	58373.6	199747	199747	199747	199747	15567	
▶ ♦ Mass Fractions												
▶ ♦ Mole Flows	kmol/hr		1676.79	1676.79	1323.77	1323.77	12450.9	12450.9	12450.9	12450.9	353.021	
▶ ♦ Mole Fractions												
▶ Molar Enthalpy	kJ/kmol		-122496	-121929	-125630	-125630	-75066.3	-75648.2	-75724.5	-77538.6	-108053	
▶ Molar Entropy	kJ/kmol-K		-345.993	-343.379	-357.856	-357.777	-110.059	-112.044	-112.296	-119.33	-289.093	
▶ Enthalpy Flow	kW		-57055.8	-56791.7	-46195.9	-46195.9	-259623	-261635	-261900	-268174	-10595.8	
▶ Average MW			44.0965	44.0965	44.0965	44.0965	16.0428	16.0428	16.0428	16.0428	44.0965	
▶ Molar Density	kmol/cum		1.32538	0.61641	12.5187	1.82009	1.29171	1.37486	1.38396	1.77024	0.135024	
▶ Heat capacity, mixture	kJ/kmol-K		103.384	97.2745	104.72	100.577	38.9101	38.8795	38.8813	40.6477	69.3549	
▶ Availability (flow), mixture	kW		-9007.51	-9106.43	-6962.85	-6971.48	-146133	-146099	-146102	-145124	-2143.59	
▶ Exergy flow rate	kW		2438.1	2336.92	2081.15	2072.46	28699.2	28745.9	28744.3	29767.6	255.774	
▶ <add properties>												

KPIs				Tab2	Tab 3
	Name	Units	Value		
▶	WTOTAL	KW	3856.16		
▶	COP	UNITLESS	1.62702		
▶	OPCOST	MM\$/OPE...	2		
▶	ACC	MM\$/OPE...	3.7		
▶	TOTCOST	MM\$/OPE...	5.7		

KPIs				Tab2	Tab 3
	Name	Units	Value		
▶	EXHGINST	MM\$/OPE...	1.5		
▶	COMPINST.1	MM\$/OPE...	5.9		
▶	DRUMINST.1	MM\$/OPE...	0.5		
▶	VLVINST.1	MM\$/OPE...	0.6		
▶	CEW.1	MM\$/OPE...	1.9		
▶	CCW.1	MM\$/OPE...	0.13		

KPIs	Tab2	Tab 3			
	Name	Units	Value		
▶	KP101	BAR	3.67		
▶	KP102	BAR	6.6		
▶	KP103	BAR	15.7449		